

Causal methods with observational data

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Research Design for Social Sciences
MA Computational Social Science, UC3M
Fall 2025

Roadmap

Housekeeping

Intro and overview

Methods in brief

Paper discussion

Re-cap and final essay

- **Groups?** Send me an email **before** **Oct 14th**

Re-cap and final essay

- Logistics:
 - 10-15 slots
 - 10min presentation + 10min feedback
 - If less groups, we can increase time
- **Questions?**

Roadmap

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Intro and overview

Methods in brief

Paper discussion

Re-cap

1. Problem/topic
2. Stories, arguments about mechanisms
3. Research question
4. Proper theory, concepts and operationalization
5. Measurement, unit of analyses, data sources, etc
6. Inference strategy
7. Results & interpretation

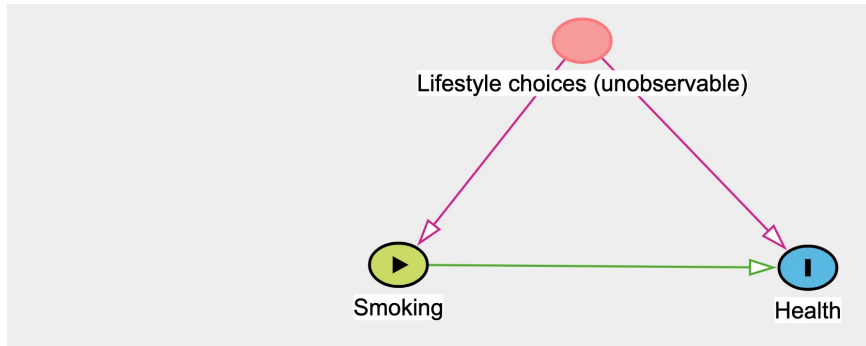
Methods and causal inference

- Most of the time is **impossible** to control for all relevant variables (i.e. not able to close all back-door paths)
- So what do we do? We try to find ways to control for unobserved confounding
- One option is to rely on **additional controlling techniques**
 - **Matching** : also depends on observables, but parametric advantages, etc (anyway, not a solution for U)
 - **Fixed effects** : can control for group-level unobservables, and correct issues related to ecological fallacy

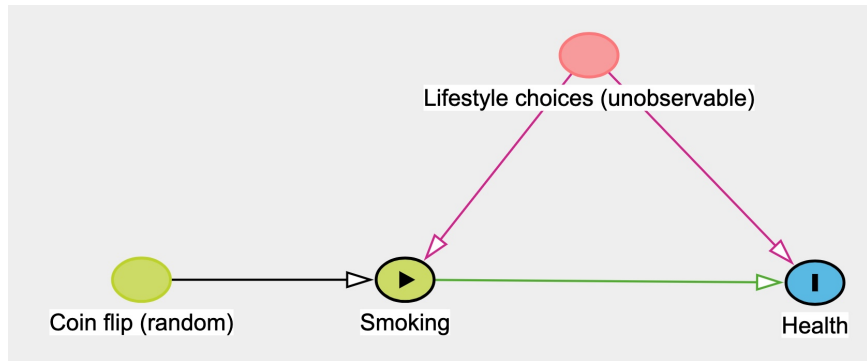
Methods and causal inference

- Another option is to find a source of **exogenous variation** and exploit
- There are a few methods related to causal inference not because they uncover causal relationships but because they are designed to exploit *'typical' exogenous sources of variation*
- Situations where there is *something* that introduces variation in the treatment that is independent from confounders and you can exploit to analyse
 - Same as randomization (of treatment assignment) in experiments
- (In H-K's *The Effect*, they're called 'template causal diagrams')

Exogenous variation



Exogenous variation



Methods and causal inference

- Five techniques commonly used in causal inference
 - two of them used for controlling (closing back-doors), and the other three to exploit 'pre-made' causal models
 - btw, what is controlling? when you control for Z , you remove the variation in X and Y that is *explained by* Z (see [this](#))

Methods and causal inference

- Five techniques commonly used in causal inference
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 - btw, what is controlling? when you control for Z , you remove the variation in X and Y that is *explained by* Z (see this)
1. Fixed effects
 2. Difference-in-differences
 3. Regression discontinuity design
 4. Instrumental variables
 5. Matching

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Intro and overview

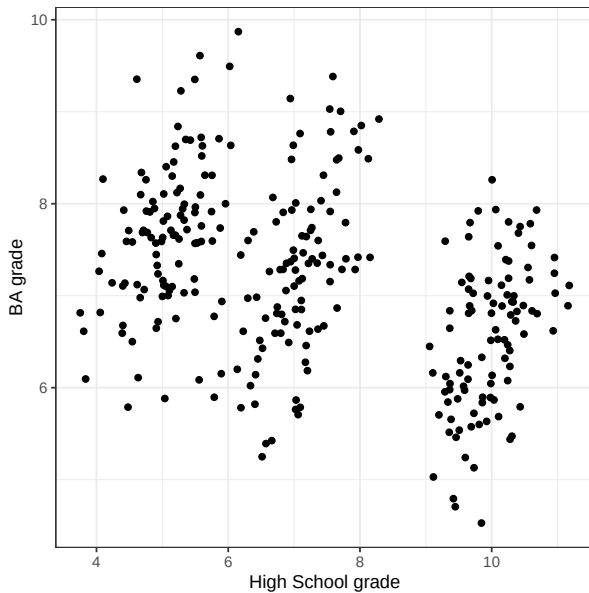
Methods in brief

Paper discussion

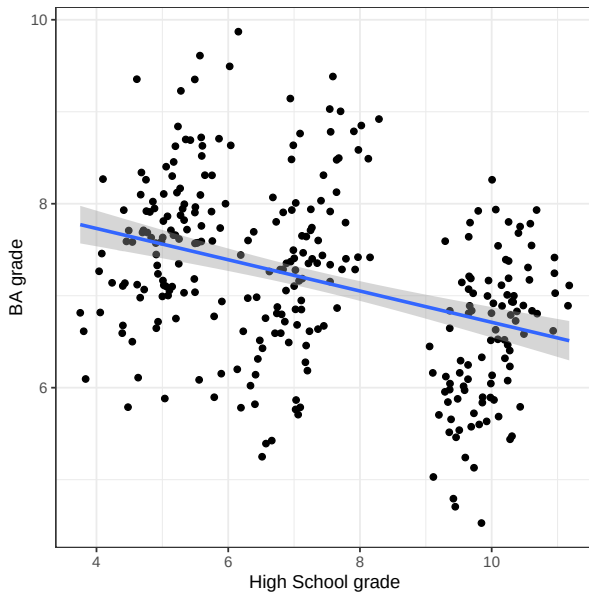
Fixed effects

- The problem with covariate adjustment is that we need to *observe* those variables, but we usually have unobserved confounders
- An strategy in that case is to try within-group comparisons, which blocks all group-wide confounding
- For example, imagine cases when our U variable is:
 - *city of origin*, in an individual-level analysis; *school effects* in an analyses of students grades; *individual background*, in a panel survey analysis...
- It corrects for **ecological fallacy** (esp. Simpson's paradox)
- Estimation: dummy variable for each group

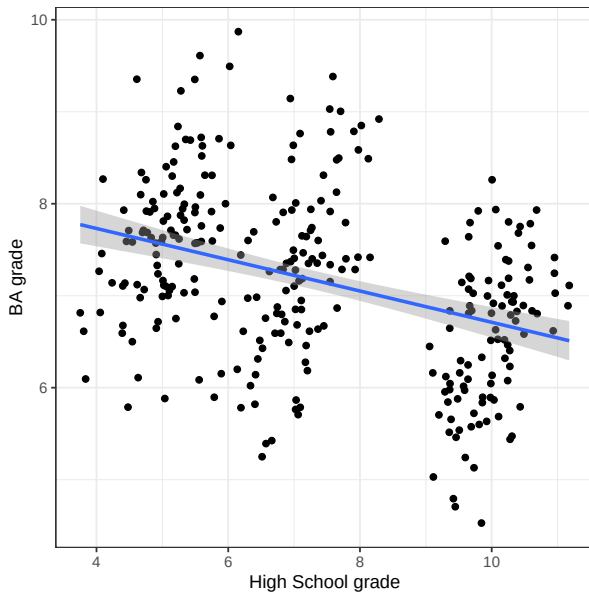
Fixed effects



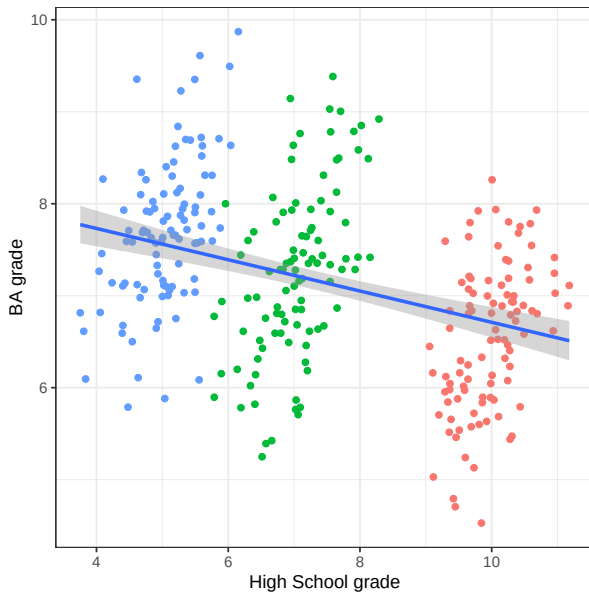
Fixed effects



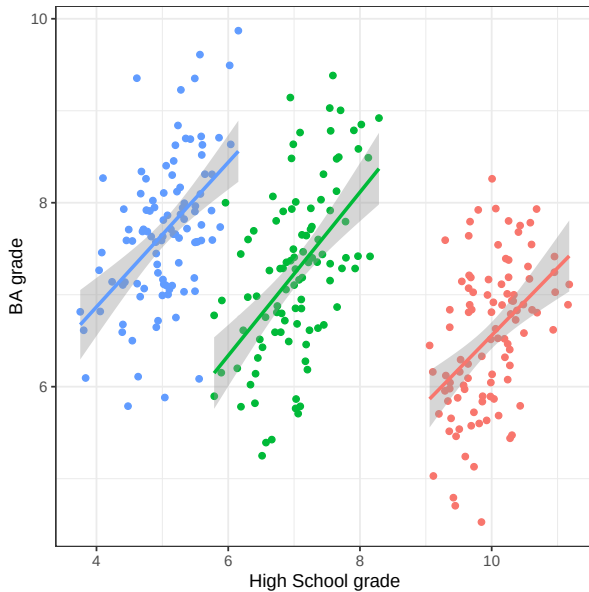
Fixed effects



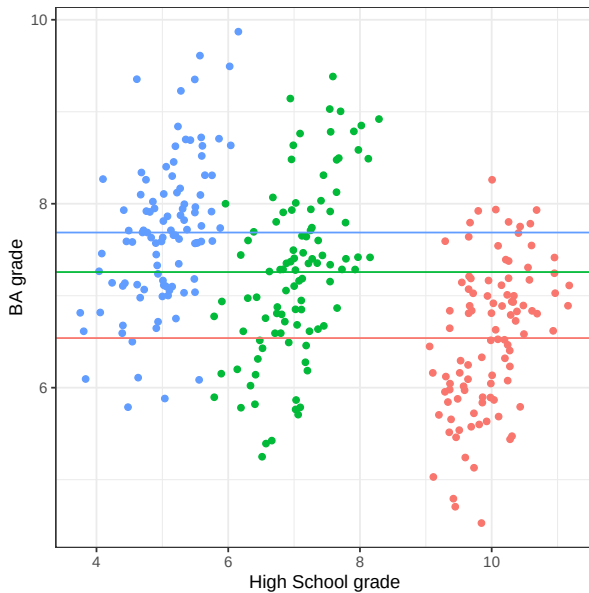
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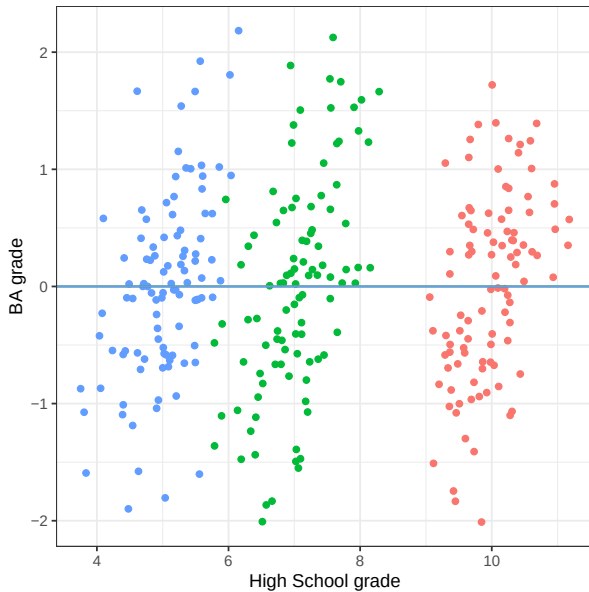
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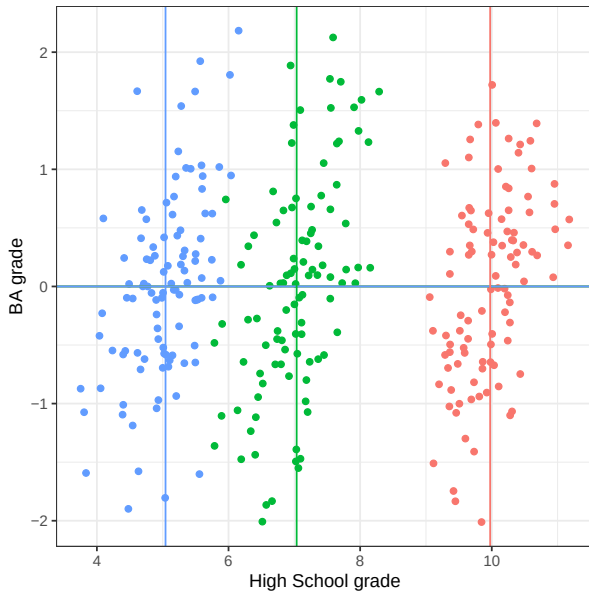
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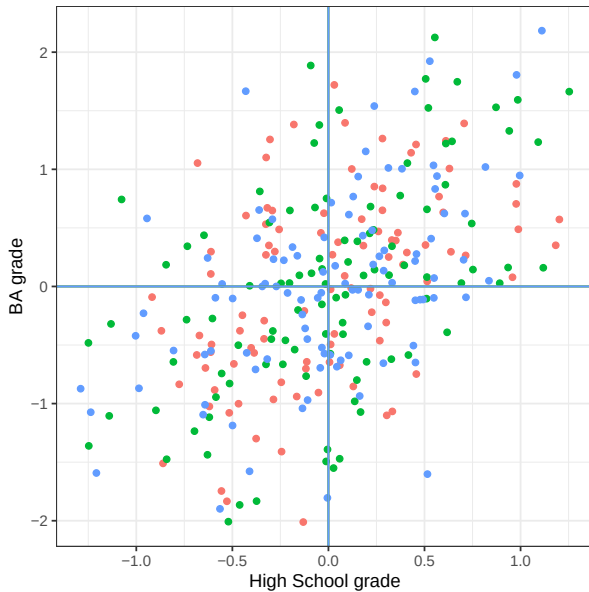
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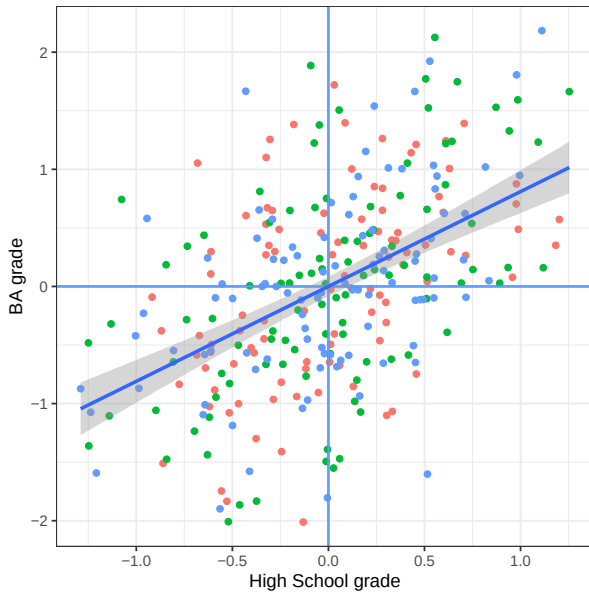
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Fixed effects



Difference-in-differences

- Treatments usually occur at a particular moment in time, e.g.:
 - Minimum wage increase, terrorist attacks, influx of refugees, ...

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- A naive idea would be to compare how it was *before* the treatment with how it is *after* the treatment, right?
 - e.g. in municipalities that removed Francoist streets, how much did Vox grow between 2016 and 2019?

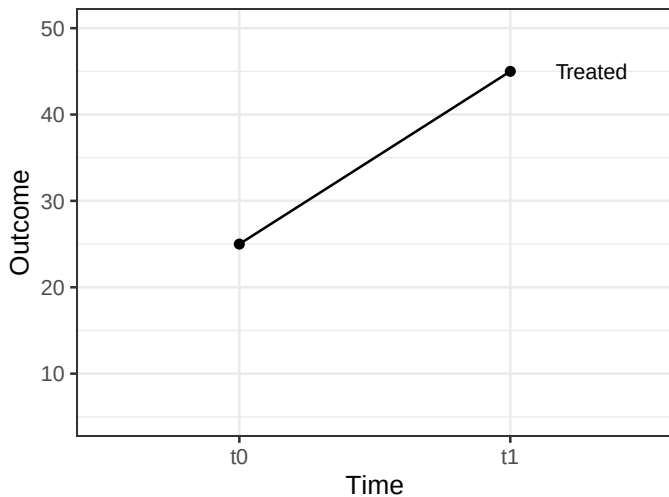
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- or we could just compare treated vs control *after* treatment
 - e.g. did Vox get more votes in 2019 in municipalities that removed F streets?

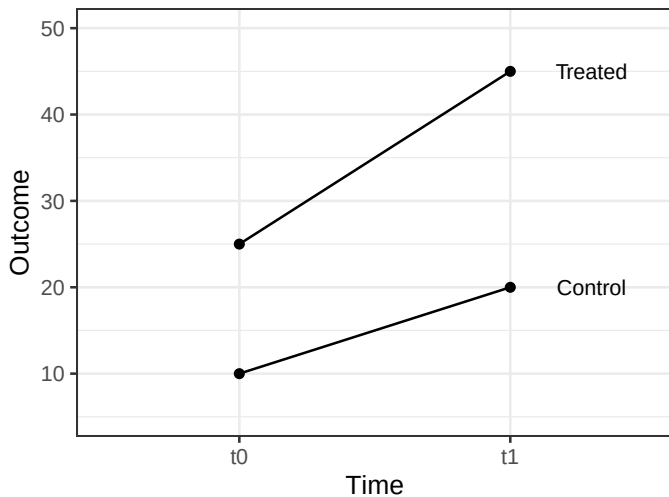
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- or we could just compare treated vs control *after* treatment
 - e.g. did Vox get more votes in 2019 in municipalities that removed F streets?
- **DiD idea**: use a control group that works as a **counterfactual** for how much would Y have changed without treatment

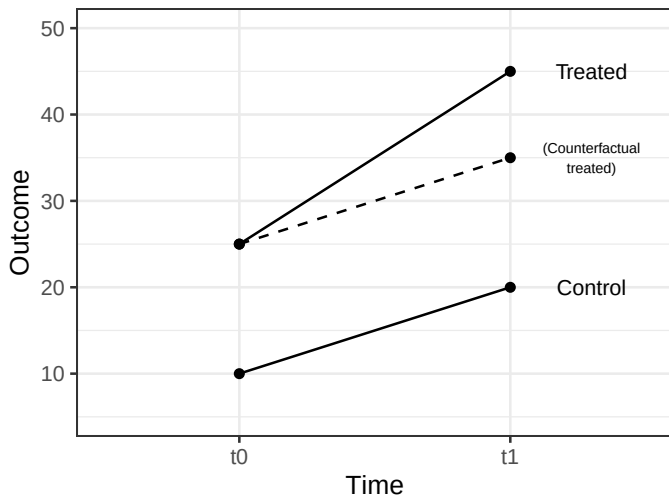
Difference-in-differences



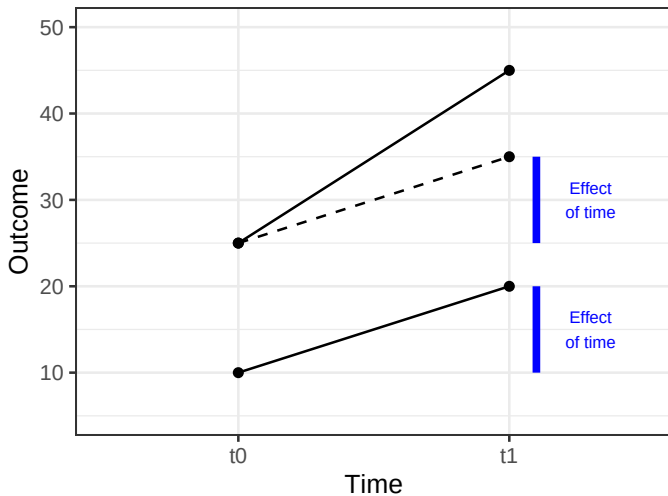
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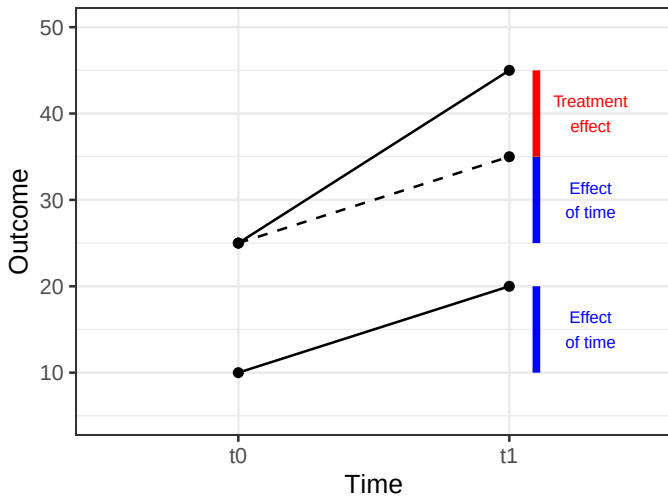
Difference-in-differences



Difference-in-differences



Difference-in-differences



Difference-in-differences estimation

- You normally estimate it using a regression model (OLS or logit, depending on the outcome) on panel data
 - each unit has one observation per period (at least, before/after treatment)
- Variables: time, group (treated or not), and their interaction

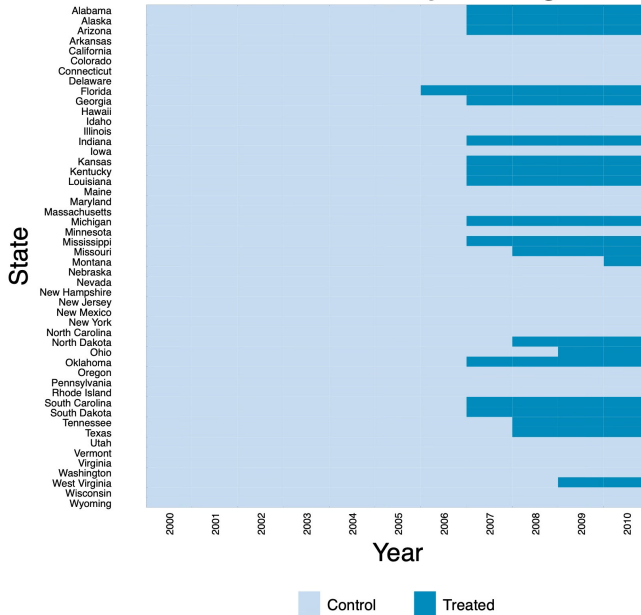
$$Y_{it} = \beta_0 + \beta_1 Treated_i + \beta_2 After_t + \beta_3 (Treated_i \times After_t) + \beta^\top \mathbf{x}_i + \epsilon_{it} \quad (1)$$

- **Key assumption**: control group is a good counterfactual
 - checking parallel trends assumption

DiD and staggered treatments

- Normal DiD works on a before/after setting, where treatment affects all units at the same time
- What if treatment is **staggered**?

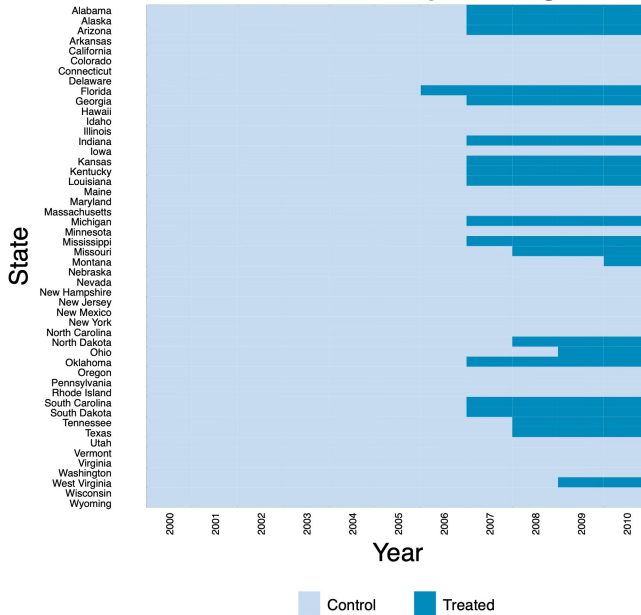
Treatment Status by Timing Group



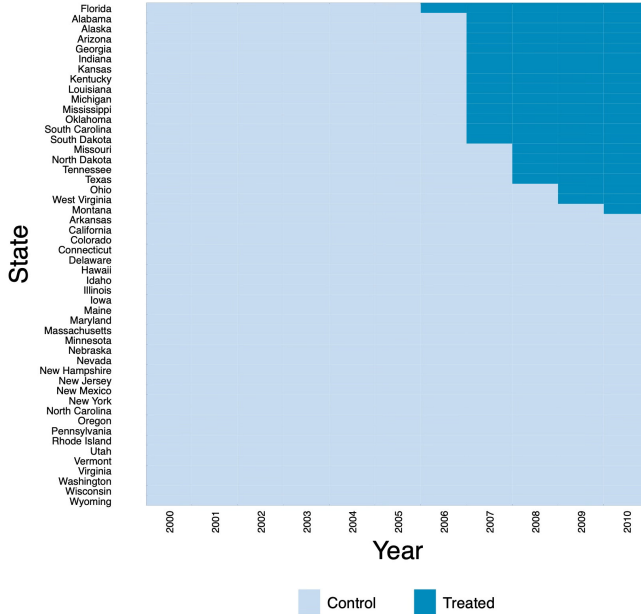
DiD and staggered treatments

- Different strategies to deal with this
- Naive: Two-Way Fixed Effects (TWFE)
 - Problem when having heterogeneous effects over time
- More recently: DiD estimates under treatment heterogeneity
 - Goodman-Bacon 2021 ("Difference-in-Differences with Variation in Treatment Timing")
 - Callaway and Sant'Anna 2021 ("Difference-in-Differences with Multiple Time Periods")
 - Roth et al. 2023 ("What's Trending in Difference-in-Differences?")

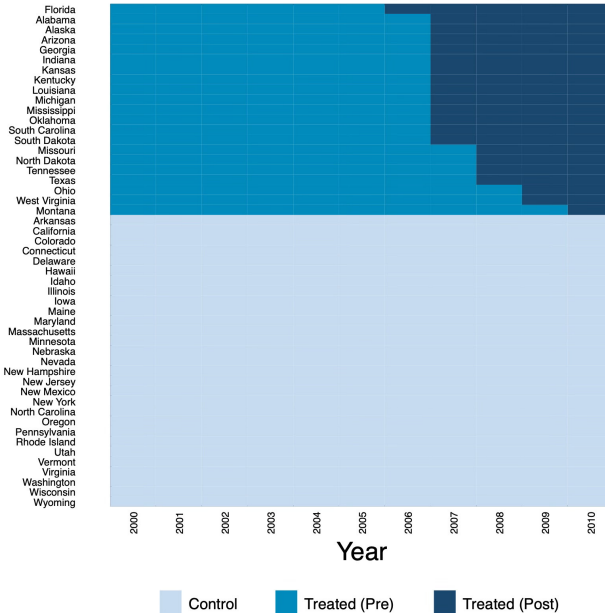
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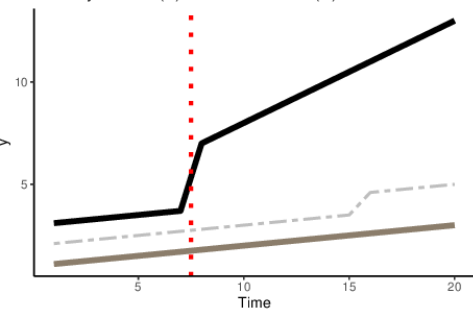
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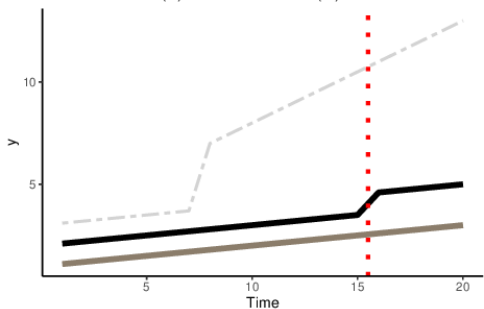
State



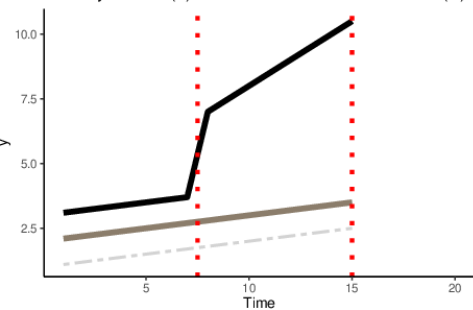
A. Early treated (T) vs never treated (C)



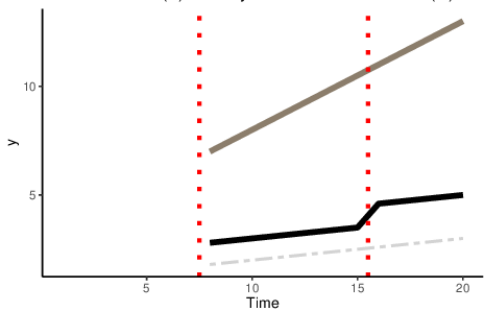
B. Late treated (T) vs never treated (C)



C. Early treated (T) vs late treated before treatment (C)



D. Late treated (T) vs early treated after treatment (C)



Regression discontinuity design

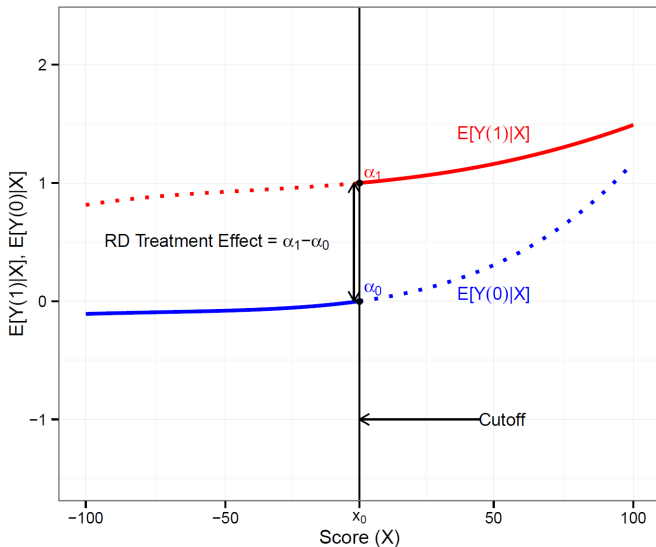
- RDD works well when assignment into treatment depends on a cutoff along a running variable
 - Do incumbent politicians have an electoral advantage? (vote share)
 - What is the effect of being drafted into the military? (birth year)
 - Effect of national policies in ethnic identification in Africa? (distance to colonial borders)

Regression discontinuity design

- RDD works well when assignment into treatment depends on a cutoff along a **running variable**
 - Do incumbent politicians have an electoral advantage? (vote share)
 - What is the effect of being drafted into the military? (birth year)
 - Effect of national policies in ethnic identification in Africa? (distance to colonial borders)
- The source of the exogenous variation:
 - Although many variables confound the relationships between X and Y , nothing should be too different *around the cutoff* between treatment and control groups (local randomization assumption)

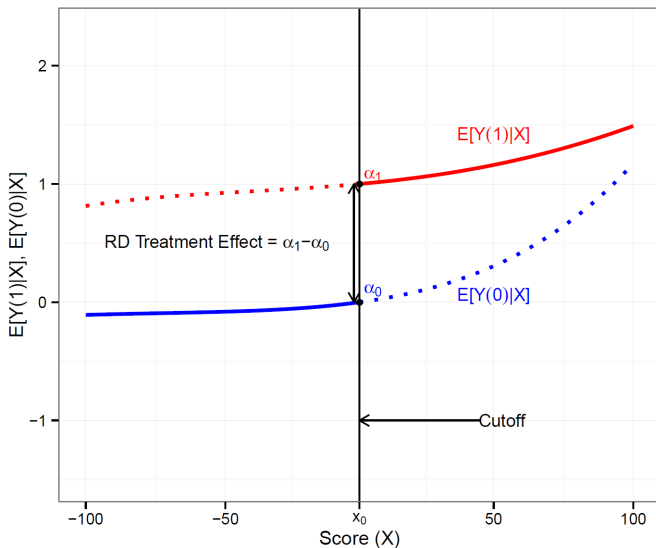
Regression discontinuity design

Figure 2: RD Treatment Effect in Sharp RD Design



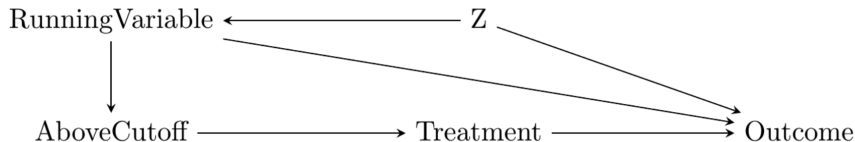
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Regression discontinuity design

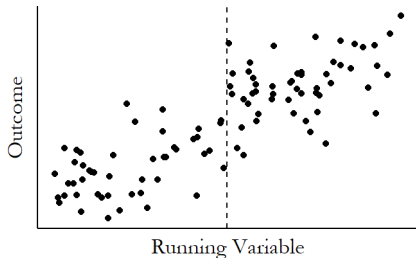
- **Key assumption**: other confounders also vary along the running variable, but are independent to the *jump over cutoff*



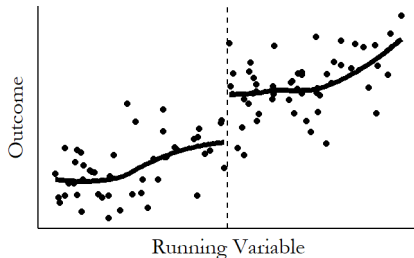
Huntington-Klein, *The Effect*, p.508

RDD estimates the **LATE**

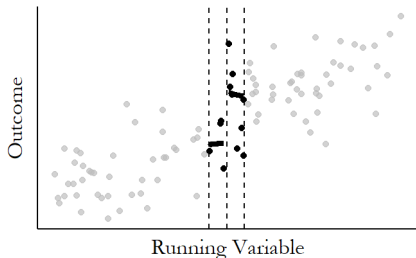
(a) Raw Data



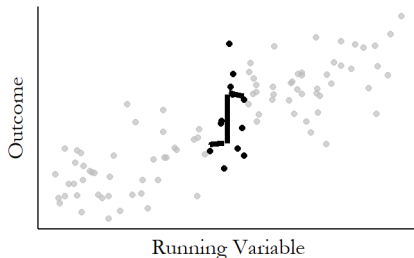
(b) Predict Values Near the Cutoff



(c) Pick a Bandwidth



(d) Estimate Jump at the Cutoff



Threats in RDD

- Outcome $\rightarrow$ cutoff: what we study might actually affect cutoff (e.g. colonial borders and ethnic ID)
 - $\rightarrow$ could also be a case of confounding, cutoff $\leftarrow Z \rightarrow$ outcome

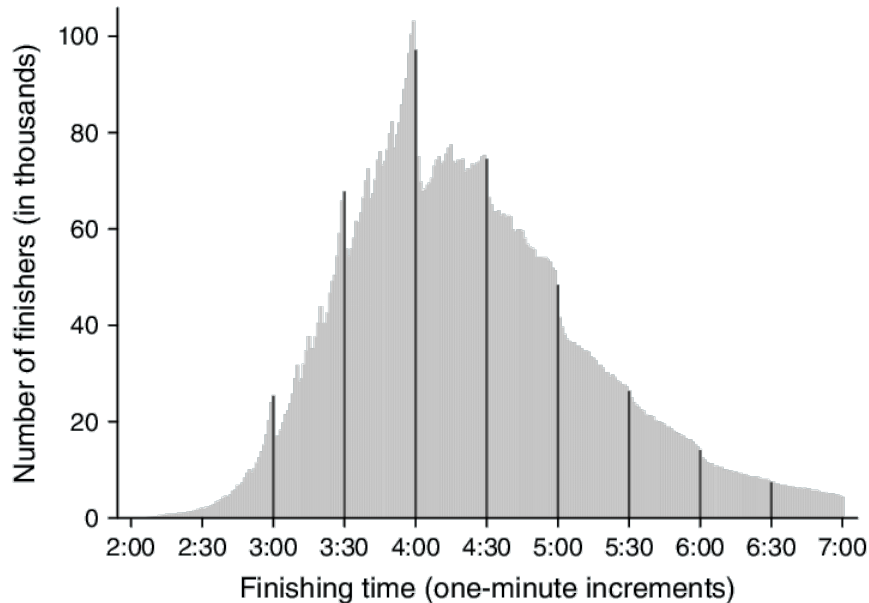
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- Precise sorting: maybe something is *actually* affecting sorting around the cutoff (next example)

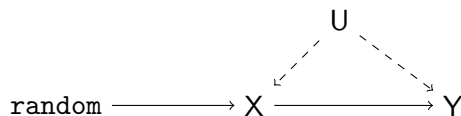
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- Precise sorting: maybe something is *actually* affecting sorting around the cutoff (next example)
- Causal effect independent of treatment: being above/below cutoff affects Y independent of the treatment we're interesting in

Threats in RDD: precise sorting



Instrumental variables



- Probably closest idea to a ‘natural experiment’

Instrumental variables

- Find an exogenous source of variation in the treatment variable
 - e.g., we want to know the effect of protests on government action, and we use *rainfall* as an instrument
- Isolate that variation and use it to identify the causal effect
 - two-stage least squares, or 2SLS

Instrumental variables

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Assumptions:

- **Relevance**: the instrument explains at least some part of the treatment variable
- Validity or **exclusion restriction**: no back door paths between the instrument and the outcome

Overview of the 3 design templates

- Difference-in-differences
- Regression discontinuity
- Instrumental variables

Example: Instrumental variables

Is Ethnic Violence Self-Perpetuating? Quasi-Experimental Evidence From Hindu-Muslim Riots in India

Journal of Conflict Resolution

2025, Vol. 0(0) 1–25

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Sam van Noort¹  **and Tanushree Goyal¹** 

Abstract

Ethnic riots tend to occur in the same places over time. We study whether this serial correlation exists because ethnic riots tend to be self-perpetuating or because both past and future riots are caused by the same underlying factors that persist through

Example: Instrumental variables

Although there is little consensus about the causes of ethnic riots, one of the most robust findings in the existing literature is that the risk of ethnic riots is substantially higher in communities that have experienced ethnic riots in the past (Horowitz 2001; Varshney 2001, 2002; Wilkinson 2009).

We ask whether this serial correlation in ethnic riots reflect a causal effect—meaning that ethnic riots in themselves make future riots more likely—or whether the serial correlation is entirely driven by other factors, which are not alleviated through rioting, and which therefore also tend to cause future riots.

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- What's the **problem** here?

Example: Instrumental variables

Studying whether ethnic riots are self-perpetuating is challenging given that the same underlying conditions that ignited riots initially may persist and therefore also cause subsequent riots, creating a classic confounding problem (Collier and Sambanis 2002; Hazlett 2019).¹ Given our imperfect knowledge of what all the relevant confounding factors are, and how to measure them sufficiently precisely, and given the fact that randomly assigning ethnic violence ourselves is both unethical and impossible, causal identification has to effectively come from a credible quasi-experiment, which accounts for confounding by design.

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- What could be a source of **exogenous variation**? (i.e. quasi-experiment)

Example: Instrumental variables

Instrument

To identify the causal effect of ethnic riots over time we follow the innovative identifications strategy of Iyer and Shrivastava (2018) by regressing future Hindu-Muslim riots on past Hindu-Muslim riots, while instrumenting the latter with whether in that same year an important Hindu festival occurred on a Friday, or not.

This identification strategy relies on five key assumptions. First, Friday festivals are correlated with Hindu-Muslim riots (relevance assumption). Second, Hindu festivals do not fall on Friday because of other factors that affect Hindu-Muslim riots, and Hindu festivals do not fall on Friday because of past Hindu-Muslim riots (exogeneity assumption). Third, Friday festivals do not affect future Hindu-Muslim riots through any other channel than present Hindu-Muslim riots (excludability assumption). Fourth, Friday festivals do not lead to a decrease in the probability of Hindu-Muslim riots in any district within our sample (monotonicity assumption). Last, Friday festivals in one area do not affect the probability of Hindu-Muslim riots in any other area within the sample (stable unit treatment value assumption).

We now discuss why we believe that these five identifying assumptions are likely to hold in our specific case.

Example: Regression discontinuity

American Political Science Review (2025) 1–21

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Political Emancipation and Modern Jewish National Identity

CARLES BOIX *Princeton University, United States, and University of Barcelona, Spain*

Following the rise of liberalism and nationalism during the nineteenth century, Jewish national identity varied across countries. While Western European and American Jews mainly came to think of themselves as nationals of their country of citizenship, a growing number of Eastern European Jews claimed to be a separate nation with a legitimate claim to self-government. Comparing the evolution of Jewish identities across North America and Europe and leveraging a regression discontinuity design based on the differential treatment of Polish and Russian Jews under Tsarism, I find that their divergent national identities responded to the extent to which Jews were politically emancipated in the country where they lived over the long century that followed the Atlantic Revolutions. Social and economic modernization played a weaker role, suggesting the need to think about national identity formation as endogenous to political and constitutional transformations marking the birth of the contemporary era.

FIGURE 1. Emancipation and Zionism

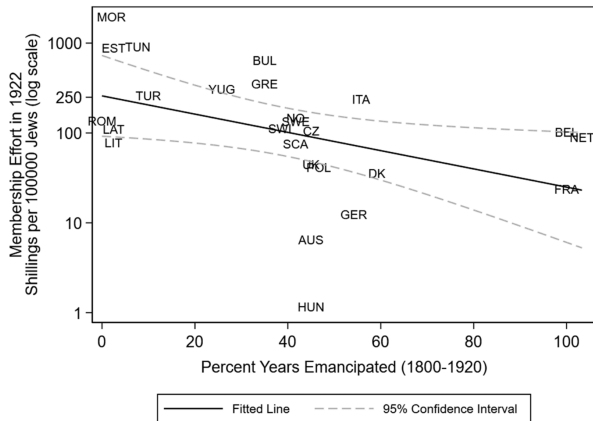


FIGURE 4. Location of Zionist Schools in Interwar Poland's Towns

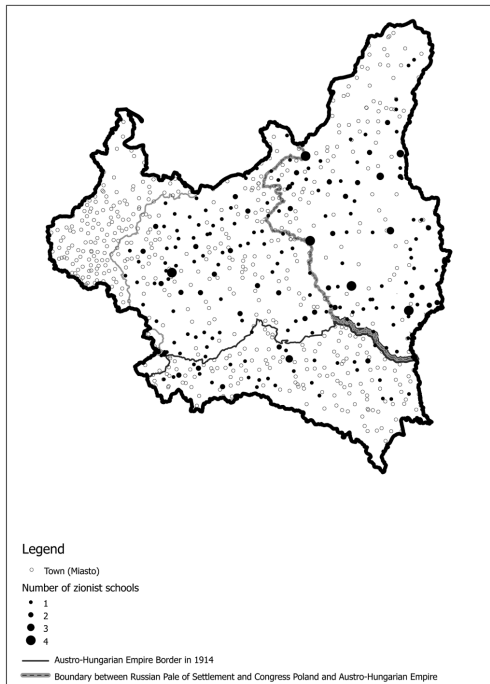


FIGURE 6. Spatial and Demographic Discontinuities at the Border

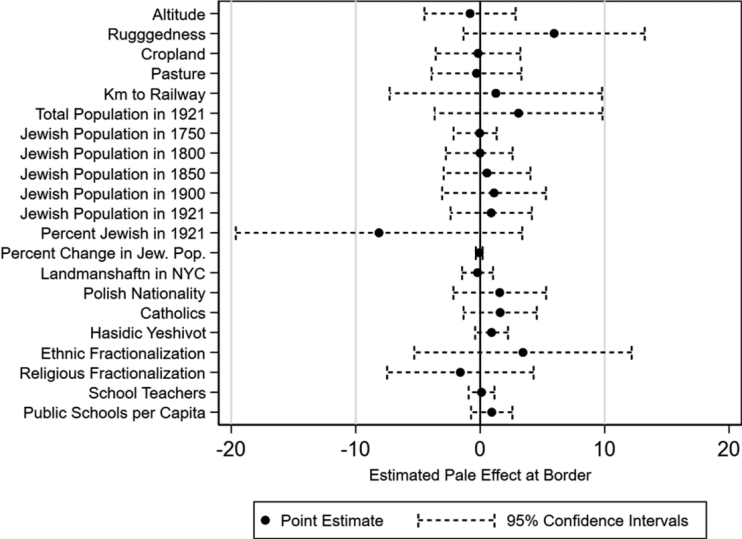
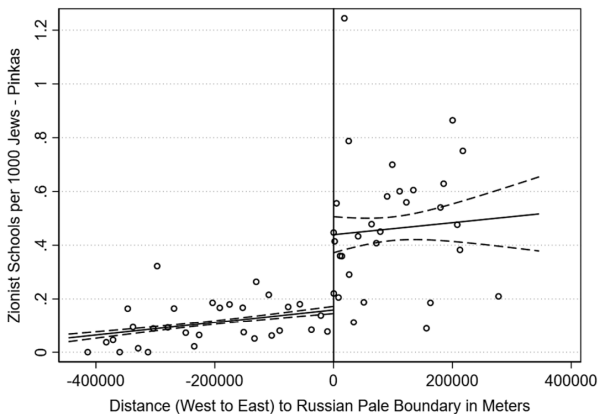


FIGURE 7. Zionist Schools in Towns



Note: Schools aggregated in bins with an equal number of observations. Source: Pinkas Ha-kehilot.

Alternative approaches to IV: build your own

Roads to Rule, Roads to Rebel: Relational State Capacity and Conflict in Africa

Carl Müller-Crepon¹ , Philipp Hunziker²,
and Lars-Erik Cederman³

Abstract

Weak state capacity is one of the most important explanations of civil conflict. Yet, current conceptualizations of state capacity typically focus only on the state while ignoring the relational nature of armed conflict. We argue that opportunities for conflict arise where relational state capacity is low, that is, where the state has less control over its subjects than its potential challengers. This occurs in ethnic groups that are poorly accessible from the state capital, but are internally highly interconnected. To test this argument, we digitize detailed African road maps and convert them into a road atlas akin to Google Maps. We measure the accessibility and internal connectedness of groups via travel times obtained from this atlas and simulate road networks for an instrumental variable design. Our findings suggest that

Journal of Conflict Resolution

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Alternative approaches to IV: build your own

Instrumental Variable Approach

We complement our robustness checks with an instrumental variable (IV) strategy that addresses potential omitted variable biases not captured by the previous tests. In particular, there might be hitherto unmeasured group-level characteristics that have affected colonial road building and recent conflict. To address such endogeneity as well as potential systematic measurement bias in the Michelin maps, our IV approach exploits variation from road networks simulated on the basis of countries' population distribution. Our IV approach improves identification by isolating the component of RSC that is due to the spatial population distribution within a country. While population distributions are less malleable than road networks, populations are not randomly distributed. We must therefore rely on the assumption that the population distributions that produce our simulated road networks are conditionally exogenous to conflict. We address potential violations of this assumption below.

Alternative approaches to IV: build your own

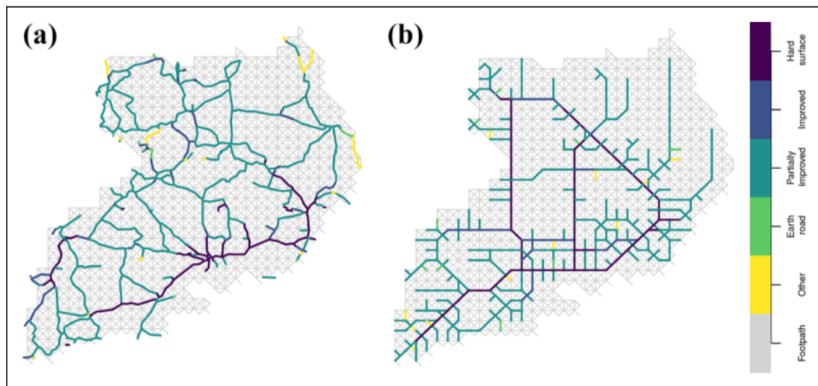


Figure 3. Observed and simulated road network in Uganda, 1966. (a) Observed network. (b) Simulated network.

(Extra) Example on theory, evidence and putting it all together

American Political Science Review (2025) 1–18

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Race, Democracy, and Public Support for War

MICHAEL TOMZ *Stanford University, United States*

JESSICA L. P. WEEKS *University of Wisconsin–Madison, United States*

Studies have found that voters in democratic countries are far more reluctant to use military force against democracies than against nondemocracies. This pattern may help explain why democracies almost never wage war against other democracies. In an important contribution, Rathbun, Parker, and Pomeroy (2025) propose that the apparent democratic peace in public opinion is an artifact of failing to account for race. Rather than democracy itself influencing support for war, they argue, the term “democracy” cues assumptions about the adversary’s racial composition, and those racialized assumptions are the true drivers of support for war. We reevaluate RPP’s evidence, concluding that their data do not support their predictions. In fact, their novel experiments provide powerful evidence that democracy affects support for war, independent of race. Our findings contribute to major debates about both regime type and race in international relations, as well as the design and interpretation of survey experiments.

Remember, usual causal inference strategies

1. Fixed effects
2. Difference-in-differences
3. Regression discontinuity design
4. Instrumental variables
5. Matching

Matching

- Main idea behind matching: adding control variables is *not the only* way to control / close back doors
- When you use a subset selecting on a variable, you are controlling for that variable
 - e.g. if you analyze $income \leftarrow education$ and only use data from individuals who grew up in big cities, you are already controlling for urban/rural

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- Example of $Y \leftarrow Treatment$ matching on X

Matching

- Normally you match treated units on a few control units (so you would get the **ATT**)
- But in some cases you do the opposite, because you have more treated units than control (and you get the **ATC**)
 - Example from Ukraine survey paper (combining matching with other)
- Two basic approaches:
 - **Distance matching:** you select observations that have similar values on the matching variables
 - **Propensity score matching:** we estimate the probability of being treated based on a set of variables, and then control/subset for that propensity score

Overall inference strategy

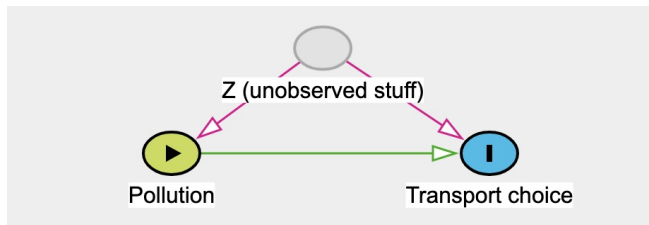
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 - We can just compare the *mean* of the treatment group and the *mean* of the control group, that's it
- In observational data, these two types of tools (controlling + exploiting exogenous variation) are often used in combination
- But **always remember** these methods **depend** on a **causal model**
- Let's look at an example (from Huntington-Klein *The Effect*, kind of)

Controlling **and** exploiting exogeneity

- Q: **Does pollution determine transport choice?**
- And say we are going to observe variation across *days*
 - Outcome: car driving
 - Treatment: pollution levels
- Clear problem of endogeneity, no? (driving in t_{-1} , economic activity, etc)



Controlling **and** exploiting exogeneity

Chasing Clean Air: Pollution-Induced Travels in China

Shuai Chen, Yuyu Chen, Ziteng Lei, Jie-Sheng Tan-Soo

Abstract: This study uses “big” data to empirically investigate a highly effective, but underexamined way of reducing one’s exposure toward air pollution—short-term travel. We determine subscribers’ locations using mobile phones’ signals and thereby establish linkages between air pollution and short-term population movements between cities in China. **Using an instrumental variable based on daily variation in wind directions and pollution levels from distant upwind locations,** we find that a one-unit increase in the origin city’s air quality index (AQI) over the destination city’s AQI increases short-term population flow from origin to destination by 0.15%. Further analyses provide richer characterizations of the decision-making process behind travel movements. Our findings add to the evidence base by examining in detail an understudied behavioral response toward air pollution.

JEL Codes: O15, Q53, Q56

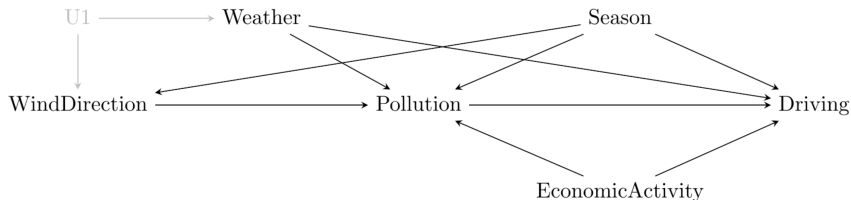
Keywords: air pollution, short-term population movement, avoidance cost, averting

Controlling **and** exploiting exogeneity

- But it's not enough with wind direction, no?

Controlling **and** exploiting exogeneity

- But it's not enough with wind direction, no?



(And city, in this case)

Controlling **and** exploiting exogeneity

- So in this case, we need to do two things:
 1. Exploit exogenous variation
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Controlling **and** exploiting exogeneity

- So in this case, we need to do two things:
 1. Exploit exogenous variation
 - In this case, using an instrumental variables approach, which isolates the variation in pollution which is explained by variation in wind direction
 2. *And* control for weather, season, and city
 - Using a regression with control variables, or matching, or fixed effects

Roadmap

Housekeeping

Intro and overview

Methods in brief

Paper discussion

Do TJ policies cause backlash? Evidence from street name changes in Spain

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Francisco Villamil¹  and Laia Balcells²

Abstract

Memories of old conflicts often shape domestic politics long after these conflicts end. Contemporary debates about past civil wars and/or repressive regimes in different parts of the world suggest that these are sensitive topics that might increase political polarization, particularly when transitional justice policies are implemented and political parties mobilize discontentment with such policies. One such policy recently debated in Spain is removing public symbols linked to a past civil war and subsequent authoritarian regime (i.e., Francoism). However, the empirical evidence on its impact is still limited. This article attempts to fill this gap by examining the political consequences of street renaming. Using a difference-in-differences approach, we show that the removal of Francoist street names has contributed to an increase of electoral support for a new far-right party, Vox, mainly at the expense of a traditional right-wing conservative party, PP. Our results suggest that revisiting the past can cause a backlash among those ideologically aligned with the perpetrator, and that some political parties can capitalize on this.

Keywords

Transitional justice, voting, conflict memories, Spain